

Weeks 2-10

K-1

Spoken instructions; matching, building, and small counts.

Grade bands are entry points. Begin with objects, then try to explain what happens.

Give one page at a time. The packet is a menu, not a checklist for one hour.

Keep facilitator solutions separate from student pages.

Week and investigation	PDF pages
02 Switches and lamps	2-3
03 Repeating secret machines	4-5
04 Around the ring	6-7
05 Tower cities	8-9
06 Code-breaking with guarantees	10-11
07 Take-away games	12-13
08 Rooks and piles	14-15
09 Bouncing paths	16-17
10 Bridges and delivery routes	18-19

Print US Letter, single-sided, at 100% / Actual Size.

Page ranges above count this cover as page 1. Bookmarks open each week.

Weekly footers retain their own page numbers and activity IDs for the use log.

Revised September 20, 2026. Planned activities; record actual use after teaching.

Two at a time

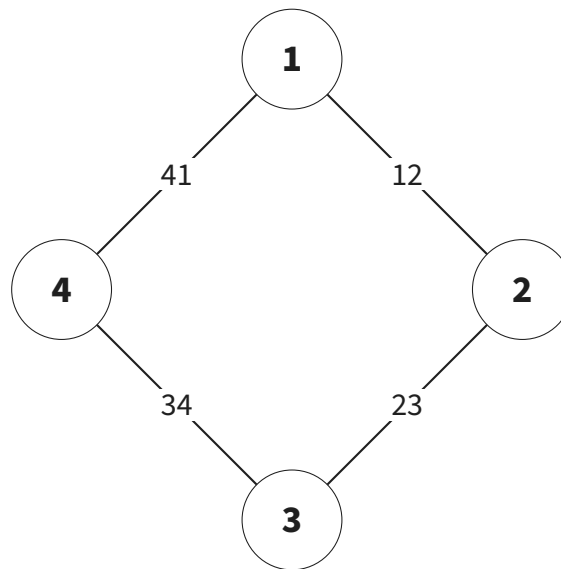
Name: _____

Date: _____

Put a two-sided counter on each lamp. White means OFF; dark means ON. One move: choose a line and **flip both counters at its ends**. Keep the counters in their places.

1. Learn the switches.

Start all OFF. Try a line. Try that same line again. What happened? Try any three moves and then undo them.



2. Make these lights.

Start all OFF each time. Can you make just the top and right lamps ON? What about just the top and bottom?

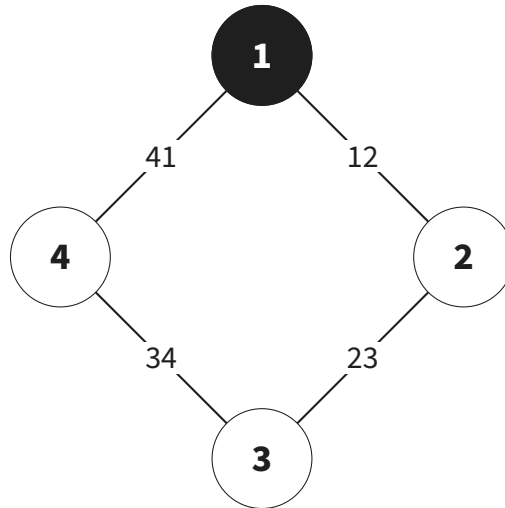
Adult: draw or write a move the child wants to remember.

Try it with objects. Explain by pointing, talking, or drawing.

A lamp that will not listen

3. Can you light just one?

Start all OFF. Try to leave only the top lamp ON. Try a few ideas, then pause and look for a reason.



A useful check: after each move, pair up the ON lamps. Is one always left over, or can they always make pairs?

4. A switch with a new rule.

Ask an adult to add a button that flips **just one** lamp. Now can you make the picture above? Which rule made the difference?

Show or tell what you noticed. An adult may record it.

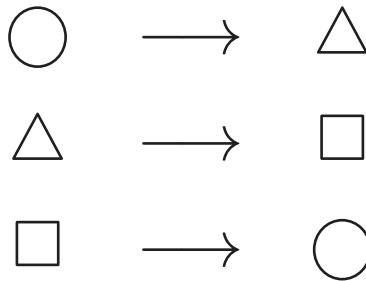
Try it with objects. Explain by pointing, talking, or drawing.

The changing-symbol machine

Name: _____

Date: _____

One turn changes each symbol by the same key. Point, draw, or use symbol cards. Try making messages before using the key.



1. Send it through.

What comes out of this message? Read one symbol at a time.



My new message (draw or place cards):

2. Run it backward.

Your friend receives □. What went in? Can you point backward along an arrow to check? Try each symbol.

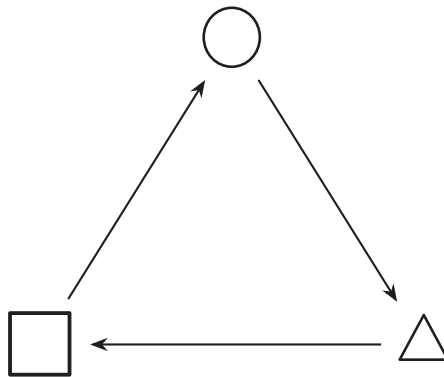
Make a message for someone else to decode.

Try it with objects. Explain by pointing, talking, or drawing.

When does it come home?

3. Keep going!

Put a counter on the circle. Follow one arrow each turn. How many turns before it first comes back? Try starting at a different symbol.



4. Change the key.

Now circle becomes triangle, triangle becomes circle, and square stays square. Will all symbols come home together after two turns? Show it with three counters.

Adult: record the child's experiment or explanation.

Try a broken key: If circle and triangle both become square, can a receiver know which one you sent?

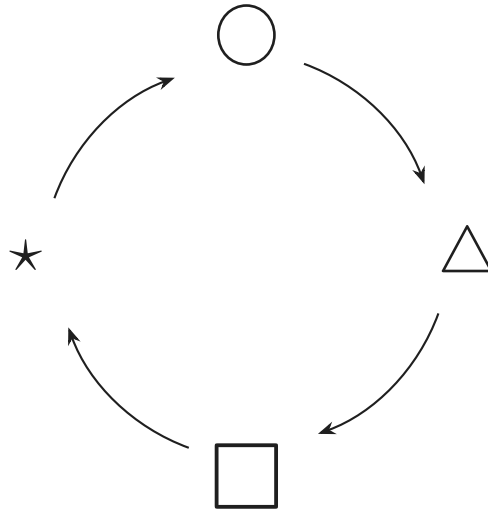
Try it with objects. Explain by pointing, talking, or drawing.

Hop around the ring

Name: _____

Date: _____

Put a counter on a shape. A hop goes to the next shape clockwise. Count the spaces **moved**, not the place where you start. Play with the ring first.



1. One hop per turn.

Start on circle. Keep taking one hop. Show when you first come home. How many turns did it take?

2. Two hops per turn.

Start on circle again. This time every turn is **two** hops. Where do you land? Can you ever land on triangle?

Circle the places you land, or tell an adult.

Try it with objects. Explain by pointing, talking, or drawing.

Find the hidden turn

3. A secret rule.

An adult moves every counter the same number of hops. You know circle becomes square. Where will triangle go? Use the ring on page 1.

Show where all four shapes go.



4. Undo the secret.

Can you find one turn that brings every counter back? Does the same turn undo itself? Test all four starting places.

My undo rule:



5. A different route.

Start on circle. Take three hops each turn. Do you visit the same places as the two-hop rule? Which rule visits every shape?

Go further: Make a ring with six picture cards. Among steps 1 through 5, which visit every card before returning home?

Try it with objects. Explain by pointing, talking, or drawing.

What can the tiny person see?

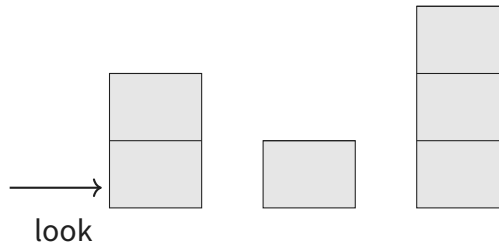
Name: _____

Date: _____

Build towers of heights 1, 2, and 3. Taller towers hide shorter towers behind them. Look from one end. Count **towers you can see**, not cubes.

1. Build and look.

Play with your towers. Then copy this row and look from both ends.



2. Same towers, different views.

Make a row showing all three towers from the left. Now make a row showing just one. Can you show exactly two?

Draw a row with two visible towers. Circle the towers you see.

3. Two ends at once.

Can you make a row that shows two towers from the left **and** two from the right? Can it show one from each end? Try; then explain.

Build first. Explain by pointing, talking, or drawing.

Have we found every skyline?

Name: _____

Date: _____

4. Find all the orders.

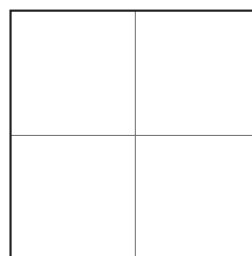
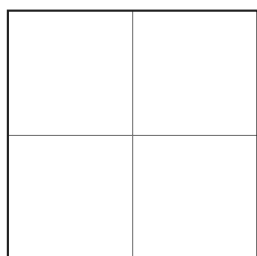
Move the towers. Have an adult record each new order. Keep the left end fixed: turning the row around may make a new order.

	Tower order	Seen left	Seen right
1			
2			
3			
4			
5			
6			

How can we be sure no order is missing?

5. A tiny city.

Use heights 1 and 2 once in **every row and every column**. Find both cities. Which one shows two towers from the left of the top row?



Build first. Explain by pointing, talking, or drawing.

Which secrets are still possible?

Name: _____

Date: _____

Two colors: R and B. Repeats allowed. A test earns one point for each color in the correct position. You hear **only the total**, not which positions match.

1. Learn the checker job.

Use two counters. First leave the secret visible. Try some tests. Check one place at a time, then say only the total. Now hide the secret and play.

2. Make all four secrets.

Build these four rows. Use different shapes as well as colors if you like.

RR

RB

BR

BB

3. One answer rules some out.

The test RR scores **1**. Cover every secret that cannot fit. How many remain?

Draw or point to the secrets that remain.

4. Choose a useful next test.

What test could tell the remaining secrets apart? Try it with both possible secrets.

My test; what its different answers would tell me:

Build first. Explain by pointing, talking, or drawing.

Make a secret-finding tree

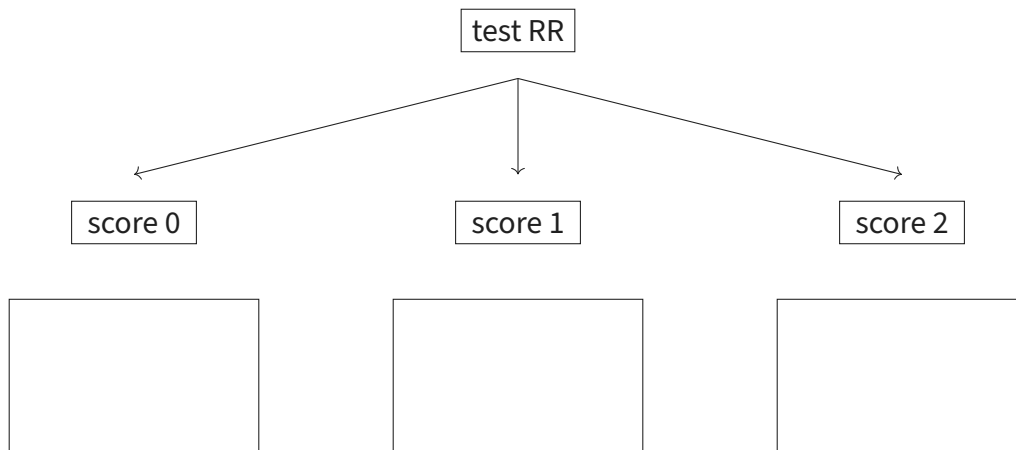
Name: _____

Date: _____

Two colors: R and B. Repeats allowed. A test earns one point for each color in the correct position. You hear **only the total**, not which positions match.

5. Begin with RR.

An adult helps record the three things you might hear. Put possible secrets below each answer.



6. Finish every branch.

If two secrets remain, choose a test that separates them. If only one remains, you can name it without another test.

Could one test always tell the secret? Show a troublesome answer.

Play again: Exchange maker and finder jobs. Can you follow your tree for every secret? An adult can do the writing; you supply the reasons.

Build first. Explain by pointing, talking, or drawing.

Can you leave a little trap?

Name: _____

Date: _____

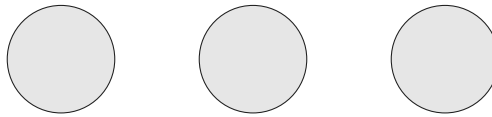
Two players alternate. Take **1 or 2** counters each turn. Never take more than remain. Taking the last counter wins.

1. Play first.

Try piles of 4 and 5 counters. Trade the first turn. Check every move.

2. A pile of three.

Choose first or second. If you choose second, show what you will do after **either** of the first player's moves.



If the first player takes 1, I will...

If the first player takes 2, I will...

3. Tell the whole story.

Does your plan work only when your partner helps, or against both choices? Demonstrate both branches with counters.

Build first. Explain by pointing, talking, or drawing.

A trap that grows

Name: _____

Date: _____

Two players alternate. Take **1 or 2** counters each turn. Never take more than remain. Taking the last counter wins.

4. Try six counters.

Arrange them as two groups of three. Can the second player answer every move and leave exactly three?



My two replies from six:

5. Choose first or second.

Try piles of 7, 8, and 9. Circle the piles you would like the **other player** to receive. Explain by playing.

7 8 9

What can I leave after the other player's turn and my reply?

Go further: Could the same plan work with three groups of three? Four groups? You can prove it by describing how one whole group disappears in each pair of turns.

Build first. Explain by pointing, talking, or drawing.

Race to the star

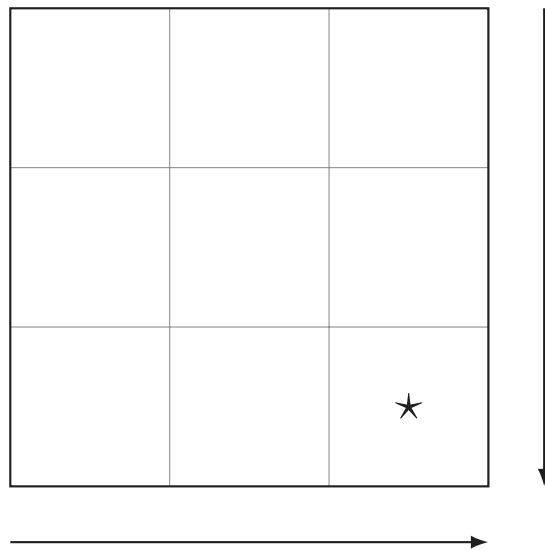
Name: _____

Date: _____

Share one token. Take turns. Slide it **right OR down** as far as you like. Move at least one square. Whoever reaches the star wins.

1. Play first.

Put a token anywhere. Show all the moves it can make. Then play three short games.



2. A tricky middle.

Start in the middle square. Can the first player win? Try **every** first move. Switch who starts.

Show what the next player can do.

Try it with objects. Explain by pointing, talking, or drawing.

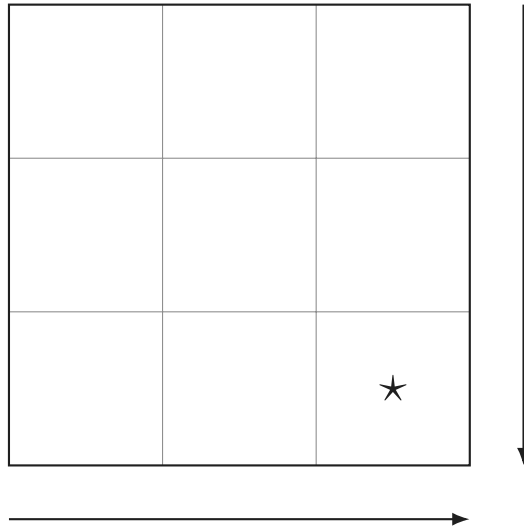
Can you copy my move?

Name: _____

Date: _____

3. Start in the top-left square.

One player moves right. Can the other copy that distance down? What if the first move goes down instead?



4. Two little piles.

Make two piles of 2 counters. Take turns taking any counters from **one pile**. Take at least one. Take the last counter to win.

Can the second player copy the first player in the other pile?



one pile



other pile

Make a challenge: Choose two piles. Would you rather go first or second? Show your reason with counters.

Try it with objects. Explain by pointing, talking, or drawing.

Bounce to a corner

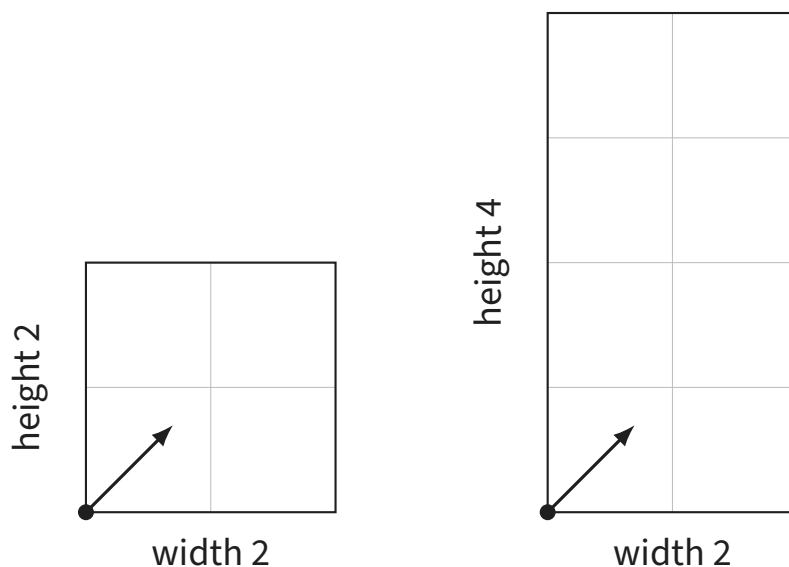
Name: _____

Date: _____

Start at the black dot. Move diagonally up-right. Bounce at a wall. Stop at the first corner. A crossing of old lines is not a wall!

1. Move a counter.

An adult draws while you choose each bounce. Try the square, then the tall table. Mark the corner where you stop.



2. Choose the next arrow.

At the **right wall**, can the ball keep going up? Which part of its direction must change? Show it with your hands.

Draw or show your bounce.

Try it with objects. Explain by pointing, talking, or drawing.

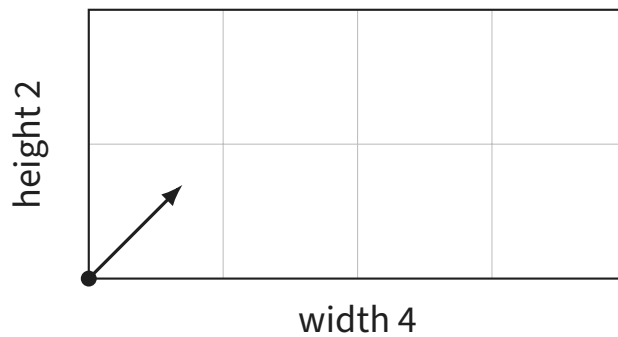
Can you see a straight path?

Name: _____

Date: _____

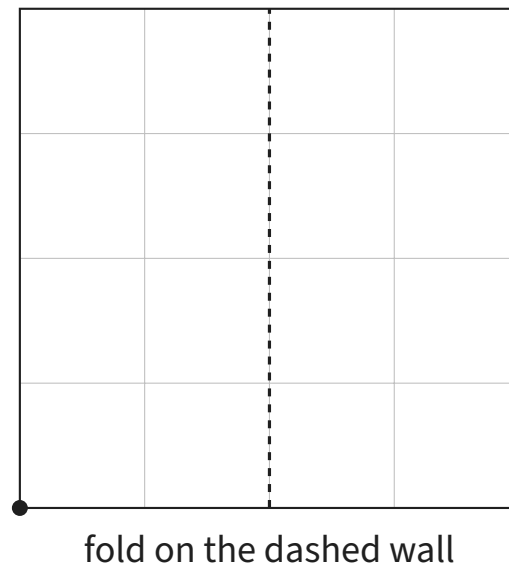
3. Try a wide table.

Will it end at the same corner as the tall table? Make a prediction, then follow the ball.



4. Open the wall.

The two rectangles below are mirror copies. Draw a **straight diagonal** from the black dot to the far corner. Fold on the middle wall. How do the two line pieces make a bounce? An adult can help fold.



Your challenge: Draw a new tall table. Predict its last corner before tracing the path.

Try it with objects. Explain by pointing, talking, or drawing.

Cross every bridge once

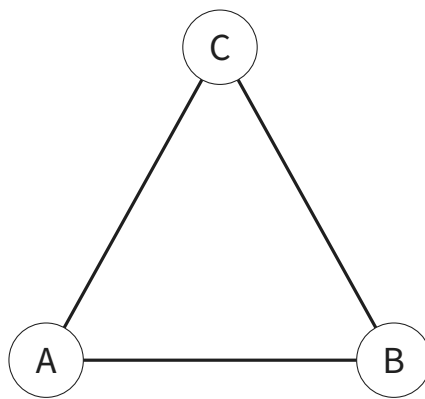
Name: _____

Date: _____

Dots are islands. Lines are bridges. Start on any island. Walk along every bridge **once**, without jumping. You may visit an island again. Put a counter on each bridge after using it.

1. Explore the island town.

Try a walk. Can you use every bridge and return to your starting island?



2. Add a little branch.

An adult draws one new bridge from A to a new island D. Can you still use every bridge once? Where can you start? Where must you finish?

Show your new map or tell an adult your route.

Try it with objects. Explain by pointing, talking, or drawing.

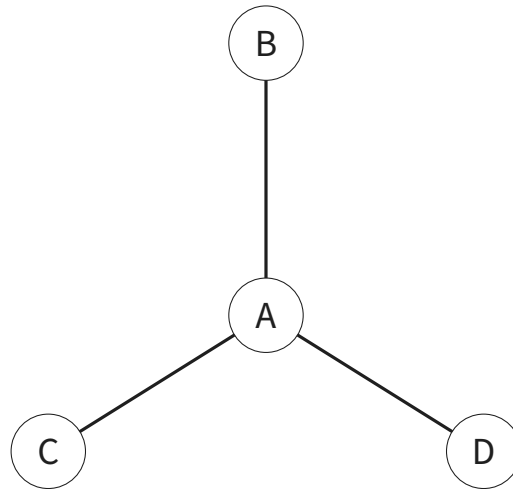
Why do dead ends matter?

Name: _____

Date: _____

3. Try this town.

Can one walk use every bridge once? Try more than one starting island.



Think with your finger: If you visit a leaf island in the middle of a walk, how can you leave without reusing its bridge?

4. Repair it.

Add **one** bridge between two leaf islands. Try your walk again. Did you need to start somewhere special?

Draw your repair and show a complete route.

Try it with objects. Explain by pointing, talking, or drawing.